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A Project Report

On

FashionPriceX- Smart Dynamic Pricing System for Fashion Products

*A Project Report submitted in partial fulfillment of the requirements for the award of
the Degree of*

Master of Computer Applications

Specialization in Artificial Intelligence & Data Science

Submitted By

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CERTIFICATE

This is to Certify that the Project work entitled “**FashionPriceX- Smart Dynamic Pricing System for Fashion Products**” is a bonafide work carried out by “**SHOBHA RANI SH**” bearing the university seat number “**24SUPMCAGL079**” in partial fulfillment for the award of Master of Computer Applications of the Sapthagiri NPS University Bengaluru during the year 2025-26. It is certified that all corrections/suggestions indicated during the review process have been incorporated in the Project report. The Project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the IV Semester Master of Computer Applications Degree.

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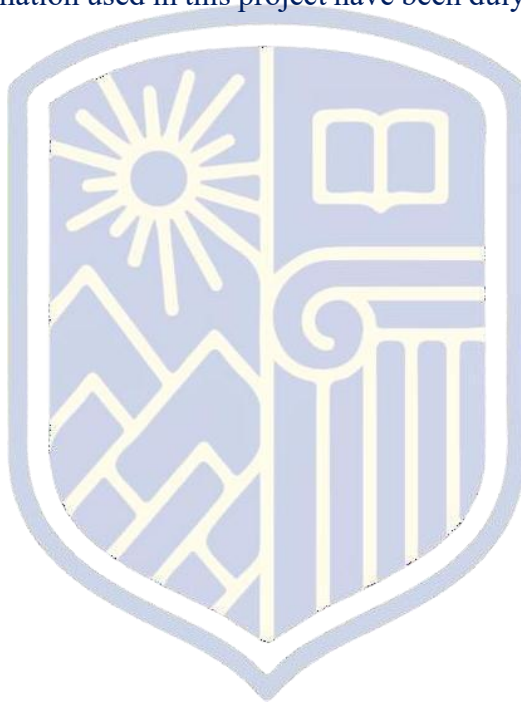
DECLARATION

I, **SHOBHA RANI SH (24SUPMCAGL079)**, a student of IV Semester Master of Computer Applications (MCA), hereby solemnly declare that the project report titled **FashionPriceX- Smart Dynamic Pricing System for Fashion Products** is a bonafide record of the original work carried out by me at the **School of Applied Science, Sapthagiri NPS University, Bengaluru**, during the **Academic year 2025–2026**.

I further declare that this project report has not been submitted, either in whole or in part, to any other university or institution for the award of any degree, diploma, or any other academic qualification. I also declare that all sources of information used in this project have been duly acknowledged and referenced.

Place: _____

Date: _____



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ABSTRACT

FashionPriceX – Smart Dynamic Pricing System for Fashion Products is an intelligent and dynamic web-based-based e-commerce application that transforms online fashion shopping experiences. Proposed FashionPriceX system differs from existing fashion websites by implementing a dynamic pricing model that leverages demand and supply forces and ensures the best prices while meeting the satisfaction of both customers and sellers. The application offers multiple interfaces including customer, seller, delivery agent, and admin panel to facilitate smart shopping, selling, order tracking, and delivery management. This project addresses the need for an intelligent and effective way of managing online shopping in the fashion industry.

Currently, most e-commerce sites and web applications focused on fashion retailing usually involve a static pricing model. Static pricing has proven to be ineffective due to the lack of flexibility and adaptation to market trends and, as a result, leads to low sales, overstocking, and reduced revenue. Therefore, a dynamic pricing model based on demand and supply principles should be implemented. The project team intended to develop a dynamic pricing web application prototype that could help maximize profits while ensuring customer satisfaction through the provision of the best prices.

FashionPriceX project is implemented using Python programming language, Flask web framework, and MYSQL as the database. The application features include User Authentication, Customer Dashboard, Seller Dashboard, Product Management, Dynamic Pricing Model, Shopping Cart, Payment Module, Order Management, Delivery Management, and Admin Dashboard among others. The project was completed following the SDLC steps including requirement analysis, design, implementation, and testing among others.

FashionPriceX innovative idea is based on utilizing a dynamic pricing model to optimize revenue, increase customer satisfaction, and ensure profitable sales. The developed web application offers a wide range of features including user authentication, shopping cart, order management, and secure payment system to facilitate shopping experiences while managing inventory effectively and profitably. In summary, the project presents a successful development of a smart and dynamic pricing system that can be used in the e-commerce industry. The project can be extended further by implementing advanced features including recommendation systems, demand forecasting models, and deploying the application on a cloud platform among others.

Keywords: Dynamic Pricing, Machine Learning, Artificial Intelligence, E-commerce, Python, Flask, MySQL, Web Application, Inventory Management.

TABLE OF CONTENTS

Ch No.	Chapters	Pg No.
1	Introduction 1.1 Introduction 1.2 Background 1.3 Problem Statement 1.4 Objectives 1.5 Scope of the Project 1.6 Existing System 1.7 Proposed System 1.8 Feasibility Study 1.9 Organization of the Report	1-3 1 1 1 2 2 2 2 2 2-3 3 3
2	Literature Survey 2.1 Introduction 2.2 Background 2.3 Review of Existing Studies 2.4 Comparative Analysis 2.5 Research Gap Analysis 2.6 Motivation for the Proposed Work 2.7 Chapter Summary	4-6 4 4 4 5 5 5 5 6
3	System Requirements 3.1 Introduction 3.2 Hardware Requirements 3.3 Software Requirements 3.4 Development Tools 3.5 Technology Stack 3.6 Software Architecture 3.7 Summary	7-8 7 7 7 8 8 8 8
4	Software Requirement Specification 4.1 Introduction 4.2 Overall Description 4.3 User Characteristics 4.4 Functional Requirements 4.5 Non-functional Requirements 4.6 Use Case Diagram 4.7 Use Case Description 4.8 External Interface Requirements	9-13 9 9 9-10 10 10-11 11 11-12 12

	4.9 Constraints	12
	4.10 Assumptions	13
	4.11 Summary	13
5	System Design	14-20
	5.1 Introduction	14
	5.2 System Architecture	14
	5.3 DFD Diagram	15
	5.4 UML Diagrams	15
	5.5 Database Design	16
	5.6 Data Dictionary	16
	5.7 Functional Design	17
	5.8 User Interface Design	17
	5.9 Sequence Diagram	18
	5.10 Activity Diagram	18
	5.11 Class Diagram	19
	5.12 Deployment Diagram	19
	5.13 Summary	20
6	Implementation	21-23
	6.1 Introduction	21
	6.2 Module-wise Implementation	21
	6.3 Class-wise Implementation (<i>if OOP-based</i>)	21
	6.4 Algorithms / Pseudocode	21-22
	6.5 Database Implementation	22
	6.6 User Interface Implementation	22
	6.7 Module Integration	23
	6.8 Security and Validation	23
	6.9 Performance Optimization	23
	6.10 Summary	24
7	Testing	24-28
	7.1 Introduction	25
	7.2 Testing Strategy	25
	7.3 Unit Testing	25-26
	7.4 Integration Testing	26
	7.5 System Testing	26
	7.6 User Acceptance Testing	26
	7.7 Test Cases	26-27
	7.8 Test Results	27

	7.9 Performance Testing	27
	7.10 Security Testing	27
	7.11 Summary	28
8	Results and Discussion	29-30
	8.1 Experimental Results	29
	8.2 Output Screens	29
	8.3 Reports Generated	29
	8.4 Performance Analysis	29-30
	8.5 Discussion	30
9	Conclusion	31
	9.1 Summary of Work	31
	9.2 Contributions	31
	9.3 Limitations	31
10	Future Scope	32
	10.1 Future Enhancements	32
	10.2 Research Directions	32
11	References	33
12	Appendices	34

LIST OF FIGURES

Fig No.	Figure Name	Pg No.
4.2.3	Use case Diagram	17
5.2.1	Represents system architecture diagram	14
5.3.1	Data Dictionary Table	16
5.4.1	Data Flow Diagram	17
5.4.2	Use Case Diagram	14
5.4.3	Sequence Diagram	18